

Research Report - Infosys Springboard Virtual Internship - 6.0

Evaluation of Models for Interactive Cyber Threat Visualization

Project: Development of Interactive Cyber Threat Visualization Dashboard

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1. Abstract

This research explores the fundamental models required to develop an interactive cyber threat visualization dashboard. In modern Security Operations Centers (SOC), data overload is a primary challenge. This report identifies and analyzes ten models categorized into threat frameworks and visualization techniques to determine their efficiency in enhancing situational awareness and reducing the Mean Time to Detect (MTTD).

2. Introduction

Cyber threat visualization is the process of transforming complex, high-velocity security data into intuitive graphical representations. An effective dashboard does not merely "show" data; it uses specific **models** to provide context, identify relationships, and suggest remediation paths. This research establishes the theoretical foundation for the project's development phase.

3. Model Research (The 10 Models)

Category A: Threat Framework & Logic Models

These models provide the "brain" for the dashboard data.

Model 1: MITRE ATT & CK Matrix

- **Overview:** A globally recognized knowledge base that catalogs adversary tactics (the "what") and techniques (the "how").
- **Dashboard Integration:** Implemented as a **Heatmap Matrix**. Cells are color-coded based on the frequency or severity of detected techniques.
- **Efficiency:** Highly efficient for mapping defensive coverage gaps and identifying specific attacker behaviors.

Model 2: Lockheed Martin Cyber Kill Chain

- **Overview:** A linear model that defines seven stages of a cyberattack, from Reconnaissance to Actions on Objectives.
- **Dashboard Integration:** Visualized as a **Progress Pipeline**. It shows at which stage a threat was intercepted.
- **Efficiency:** Efficient for executive reporting and high-level strategy, as it simplifies complex intrusions into a clear timeline.

Model 3: The Diamond Model of Intrusion Analysis

- **Overview:** Analyzes events by linking four vertices: **Adversary, Capability, Infrastructure, and Victim**.
- **Dashboard Integration:** Represented as a **Relational Graph**. When an analyst clicks an IP, the dashboard "pivots" to show the associated malware or victim.
- **Efficiency:** Exceptional for **Threat Intelligence** and attribution (understanding *who* is attacking).

Model 4: STRIDE Threat Model

- **Overview:** Developed by Microsoft, it categorizes threats into: Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege.
- **Dashboard Integration:** Used as a **Categorization Filter** for alerts, allowing analysts to sort by threat type.
- **Efficiency:** High efficiency in the "Design Phase" of security to ensure all attack surfaces are monitored.

Model 5: DREAD Risk Assessment Model

- **Overview:** A mathematical model used to prioritize threats based on: Damage, Reproducibility, Exploitability, Affected Users, and Discoverability.
- **Dashboard Integration:** Used to calculate a **Priority Score** (\$P\$) for every alert:

$$P = \frac{D+R+E+A+D}{5}$$
- **Efficiency:** Prevents "Alert Fatigue" by ensuring analysts only see high-impact threats first.

Category B: Interactive Visualization Models

These models provide the "eyes" for the user interface.

Model 6: Force-Directed Graphs (Node-Link Diagrams)

- **Overview:** Represents entities (Servers, IPs) as nodes and their interactions as links.
- **Interactivity:** Users can drag nodes, zoom, and click to reveal metadata.
- **Efficiency:** Best for detecting **Lateral Movement** (how an infection spreads from one computer to another).

Model 7: Sankey Diagrams (Flow Representation)

- **Overview:** A flow diagram where the width of the arrows is proportional to the flow quantity.
- **Dashboard Integration:** Visualizing network traffic volume.
- **Efficiency:** Highly efficient for spotting **Data Exfiltration**. A sudden "thick" flow to an unknown external IP is a clear red flag.

Model 8: Geospatial Threat Mapping

- **Overview:** Projecting IP-based threat origins onto a 2D or 3D world map.
- **Interactivity:** Time based animation showing attack "pulses" across the globe.
- **Efficiency:** Critical for **Global SOC Operations** to identify regional attack campaigns or geopolitical threat surges.

Model 9: Parallel Coordinates Model

- **Overview:** Visualizes multivariate data (e.g., Time, Port, Byte Size, Protocol) as vertical axes.
- **Interactivity:** Analysts can "brush" or filter along an axis to isolate specific traffic patterns.
- **Efficiency:** Best for **Anomaly Detection**. It helps find "outlier" packets that don't follow normal behavioral paths across multiple variables.

Model 10: Treemapping (Hierarchical View)

- **Overview:** Uses nested rectangles to represent the hierarchy of a network (e.g., Department > Server > Application).
- **Dashboard Integration:** Size indicates asset value; color (Green/Yellow/Red) indicates current threat level.
- **Efficiency:** Allows an analyst to monitor the health of **thousands of assets** on a single screen without scrolling.

4. Comparative Analysis Table

Model	Primary Use	Interactivity	Target Metric
MITRE ATT&CK	Behavior Mapping	High (Drill-down)	Technique Coverage
Kill Chain	Attack Lifecycle	Low (Static)	Interception Stage
Sankey	Traffic Analysis	Medium (Hover)	Volume / Exfiltration
Parallel Coords	Anomaly Hunting	High (Filtering)	Multivariate Correlation
Geospatial	Global Monitoring	Medium (Zoom)	Origin/Destination

5. Conclusion & Recommendation

For the "Interactive Cyber Threat Visualization Dashboard," a hybrid model is recommended. The **MITRE ATT&CK Matrix** should serve as the logical foundation for data categorization, while **Force-Directed Graphs** and **Sankey Diagrams** should be used for the primary interactive UI components. This combination ensures both strategic depth and tactical speed.

6. References

1. MITRE ATT&CK. (2025). *Enterprise Matrix Knowledge Base*.
2. Lockheed Martin. (2024). *The Cyber Kill Chain Framework*.
3. Tufte, E. (2001). *The Visual Display of Quantitative Information*.
4. NIST Special Publication 800-115. *Technical Guide to Information Security Testing*.